

10. Write a Python Program to Implement Expectation & Maximization Algorithm

a)

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_1", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
model = GaussianMixture(n_components=2, random_state=42)
model.fit(X)
clusters = model.predict(X)
print("\nCluster Assignments:")
print(clusters)
print("\nCluster Means:")
print(model.means_)
print("\nCluster Probabilities:")
print(model.predict_proba(X))
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=clusters)
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("Expectation Maximization - Dataset 1")
plt.show()
```

OUTPUT:

Dataset:

	CGPA	Aptitude Score	Communication Score	Placement
0	9.2	90	88	Placed
1	8.8	85	82	Placed
2	8.5	80	79	Placed
3	7.2	72	70	Not Placed
4	6.8	68	65	Not Placed
5	9.0	92	90	Placed
6	7.5	75	74	Not Placed
7	8.6	84	81	Placed
8	6.5	60	58	Not Placed
9	8.9	88	86	Placed

Cluster Assignments:

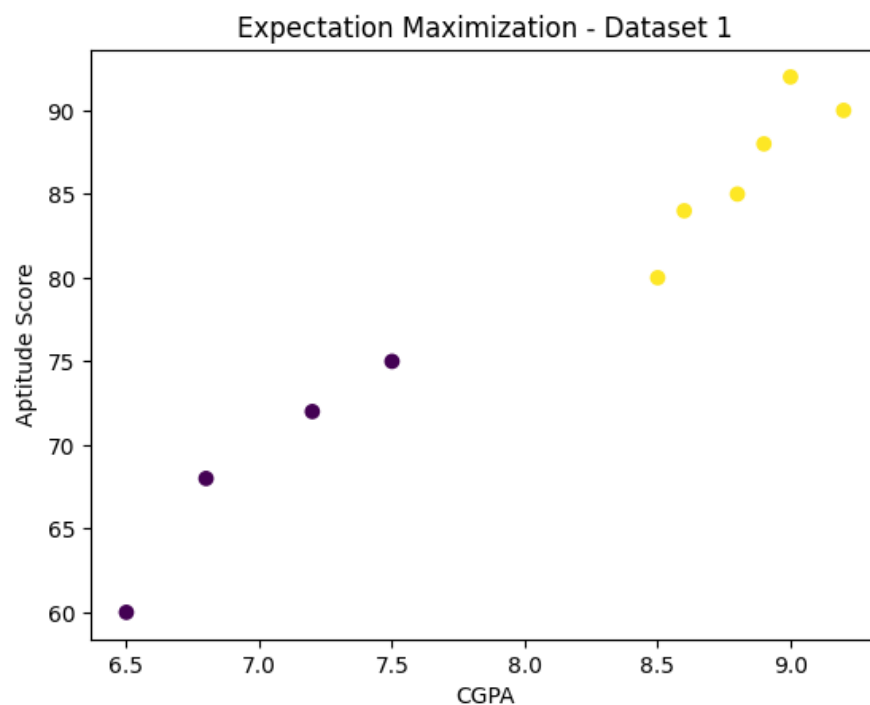
[1 1 1 0 0 1 0 1 0 1]

Cluster Means:

```
[[ 7.          68.75      66.75      ]
 [ 8.83333333 86.5       84.33333333]]
```

Cluster Probabilities:

```
[[0.00000000e+00 1.00000000e+00]
 [0.00000000e+00 1.00000000e+00]
 [0.00000000e+00 1.00000000e+00]
 [1.00000000e+00 9.95509734e-20]
 [1.00000000e+00 7.27311286e-29]
 [0.00000000e+00 1.00000000e+00]
 [1.00000000e+00 5.18858148e-15]
 [0.00000000e+00 1.00000000e+00]
 [1.00000000e+00 1.55815399e-28]
 [0.00000000e+00 1.00000000e+00]]
```



b)

Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_2", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
model = GaussianMixture(n_components=2, random_state=42)
model.fit(X)
clusters = model.predict(X)
print("\nCluster Assignments:")
print(clusters)
print("\nCluster Means:")
print(model.means_)
print("\nCluster Probabilities:")
print(model.predict_proba(X))
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=clusters)
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("Expectation Maximization - Dataset 2")
plt.show()
```

OUTPUT:

Dataset:

	Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
0	8	780	5	Yes
1	7	760	4	Yes
2	6	720	6	Yes
3	4	620	8	No
4	3	580	10	No
5	9	800	4	Yes
6	5	650	7	No
7	8	770	5	Yes
8	4	600	9	No
9	7	740	6	Yes

Cluster Assignments:

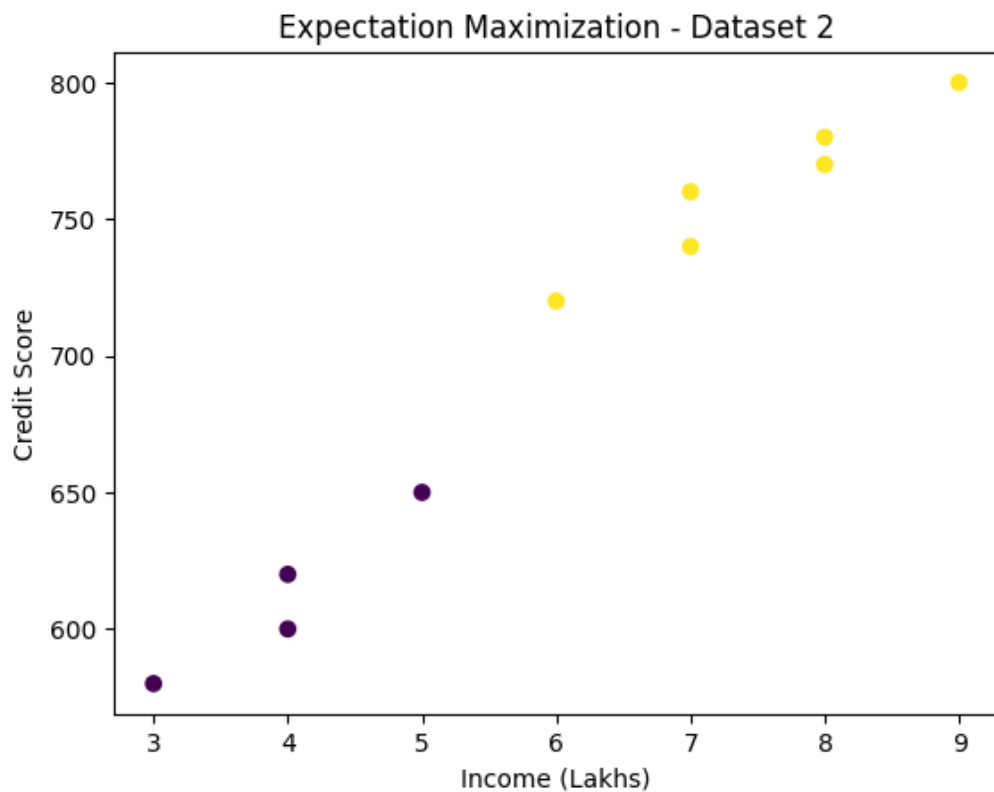
[1 1 1 0 0 1 0 1 0 1]

Cluster Means:

```
[[ 4.      612.5      8.5      ]
 [ 7.5     761.66666667  5.      ]]
```

Cluster Probabilities:

```
[[9.41933173e-226 1.00000000e+000]
 [4.47065284e-066 1.00000000e+000]
 [2.27369292e-080 1.00000000e+000]
 [1.00000000e+000 3.39124555e-031]
 [1.00000000e+000 4.53306960e-037]
 [8.84862346e-209 1.00000000e+000]
 [1.00000000e+000 7.16794142e-030]
 [2.91959802e-176 1.00000000e+000]
 [1.00000000e+000 2.30818249e-050]
 [1.64529094e-146 1.00000000e+000]]
```



c)

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39	110	94	Positive
38.5	108	95	Positive
37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_3", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
model = GaussianMixture(n_components=2, random_state=42)
model.fit(X)
clusters = model.predict(X)
print("\nCluster Assignments:")
print(clusters)
print("\nCluster Means:")
print(model.means_)
print("\nCluster Probabilities:")
print(model.predict_proba(X))
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=clusters)
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("Expectation Maximization - Dataset 3")
plt.show()
```

OUTPUT:

Dataset:

↑

↓

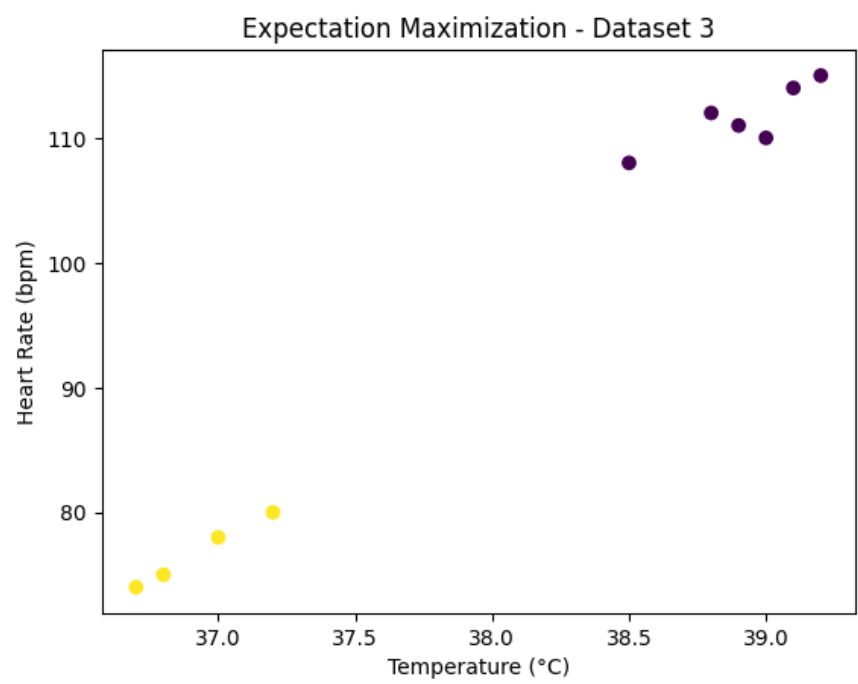
✎

	Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
0	39.0	110	94	Positive
1	38.5	108	95	Positive
2	37.0	78	99	Negative
3	39.2	115	93	Positive
4	36.8	75	98	Negative
5	38.8	112	94	Positive
6	37.2	80	98	Negative
7	39.1	114	92	Positive
8	36.7	74	99	Negative
9	38.9	111	93	Positive

Cluster Assignments:
[0 0 1 0 1 0 1 0 1 0]

Cluster Means:
[[38.91666667 111.66666667 93.5]
[36.925 76.75 98.5]]

Cluster Probabilities:
[[1.00000000e+00 0.00000000e+00]
[1.00000000e+00 0.00000000e+00]
[2.27137219e-70 1.00000000e+00]
[1.00000000e+00 0.00000000e+00]
[1.50070158e-95 1.00000000e+00]
[1.00000000e+00 0.00000000e+00]
[3.94484344e-68 1.00000000e+00]
[1.00000000e+00 0.00000000e+00]
[4.62624694e-92 1.00000000e+00]
[1.00000000e+00 0.00000000e+00]]



d)

Experience	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.mixture import GaussianMixture
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_4", nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
model = GaussianMixture(n_components=2, random_state=42)
model.fit(X)
clusters = model.predict(X)
print("\nCluster Assignments:")
print(clusters)
print("\nCluster Means:")
print(model.means_)
print("\nCluster Probabilities:")
print(model.predict_proba(X))
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=clusters)
plt.xlabel(X.columns[0])
plt.ylabel(X.columns[1])
plt.title("Expectation Maximization - Dataset 4")
plt.show()
```

OUTPUT:

Dataset:

	Experience	Performance Score	Training Hours	Promotion
0	10	95	50	Yes
1	8	90	45	Yes
2	7	88	40	Yes
3	3	65	20	No
4	2	60	18	No
5	9	93	48	Yes
6	4	70	25	No
7	8	89	42	Yes
8	2	58	15	No
9	6	85	38	Yes

Cluster Assignments:

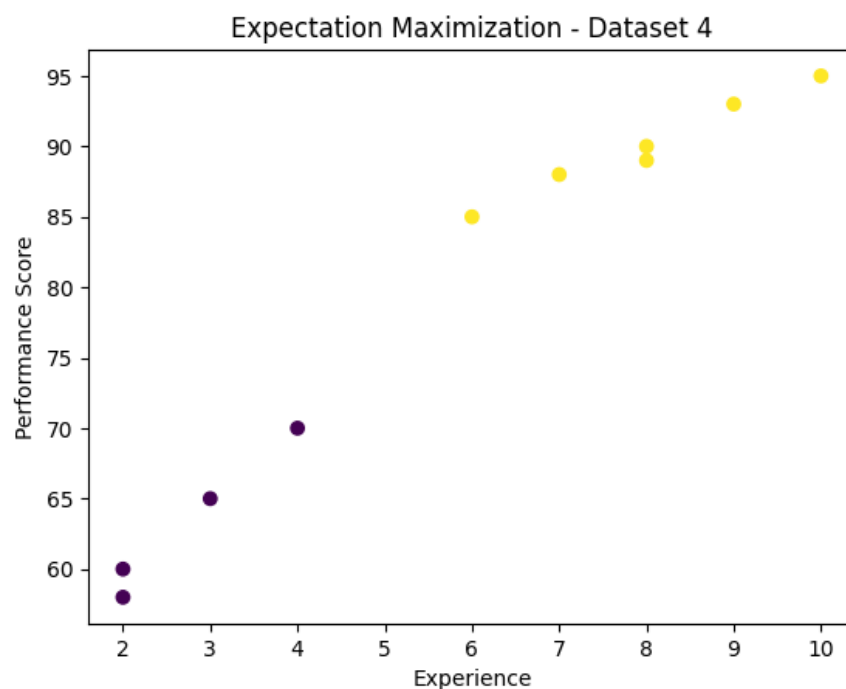
```
[1 1 1 0 0 1 0 1 0 1]
```

Cluster Means:

```
[[ 2.75000057 63.25000331 19.50000274]
 [ 8.00000009 90.00000018 43.83333368]]
```

Cluster Probabilities:

```
[[3.39296970e-112 1.00000000e+000]
 [1.08670596e-035 1.00000000e+000]
 [5.30679595e-007 9.99999469e-001]
 [1.00000000e+000 5.41040319e-166]
 [1.00000000e+000 5.96484043e-262]
 [3.73262935e-064 1.00000000e+000]
 [1.00000000e+000 2.08525177e-107]
 [4.36867807e-026 1.00000000e+000]
 [1.00000000e+000 1.70395633e-314]
 [4.87636639e-009 9.99999995e-001]]
```



e)

Weight	Diameter	Sweetness	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.mixture import GaussianMixture

df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_5", nrows=10)

print("Dataset:")

print(df)

X = df.iloc[:, :-1]

model = GaussianMixture(n_components=2, random_state=42)

model.fit(X)

clusters = model.predict(X)

print("\nCluster Assignments:")

print(clusters)

print("\nCluster Means:")

print(model.means_)

print("\nCluster Probabilities:")

print(model.predict_proba(X))

plt.scatter(X.iloc[:, 0], X.iloc[:, 1], c=clusters)

plt.xlabel(X.columns[0])

plt.ylabel(X.columns[1])

plt.title("Expectation Maximization - Dataset 5")
```

```
plt.show()
```

OUTPUT:

Dataset:

	Weight	Diameter	Sweetness	Fruit
0	150	7.5	90	Apple
1	160	7.8	88	Apple
2	140	7.2	91	Apple
3	120	5.5	70	Orange
4	125	5.8	72	Orange
5	130	6.0	74	Orange
6	155	7.6	89	Apple
7	118	5.4	69	Orange
8	148	7.4	92	Apple
9	122	5.7	71	Orange

Cluster Assignments:

```
[1 1 1 0 0 0 1 0 1 0]
```

Cluster Means:

```
[[123.    5.68  71.2 ]  
 [150.6   7.5   90.  ]]
```

Cluster Probabilities:

```
[[0.00000000e+00 1.00000000e+00]  
 [1.47777233e-72 1.00000000e+00]  
 [0.00000000e+00 1.00000000e+00]  
 [1.00000000e+00 0.00000000e+00]  
 [1.00000000e+00 0.00000000e+00]  
 [1.00000000e+00 0.00000000e+00]  
 [0.00000000e+00 1.00000000e+00]  
 [1.00000000e+00 0.00000000e+00]  
 [0.00000000e+00 1.00000000e+00]  
 [1.00000000e+00 0.00000000e+00]]
```

